Breaker Size Chart

Every standard breaker rating from 15 A to 1,200 A, the continuous load it may carry, and what that load is in kilowatts.

A standard breaker may carry a continuous load of 80 percent of its rating. That is the 125 percent sizing rule read backwards: a 20 A breaker takes 16 A of continuous load, a 100 A breaker takes 80 A. A continuous load is one expected to run at its maximum for three hours or more.

Standard ratings and continuous load

Breaker (A)	Continuous load (A)	kW at 120 V 1ϕ	kW at 240 V 1ϕ	kW at 208 V 3ϕ	kW at 480 V 3ϕ
15	12	1.44	2.88	4.32	9.98
20	16	1.92	3.84	5.76	13.3
25	20	2.40	4.80	7.21	16.6
30	24	2.88	5.76	8.65	20.0
35	28	3.36	6.72	10.1	23.3
40	32	3.84	7.68	11.5	26.6
45	36	4.32	8.64	13.0	29.9
50	40	4.80	9.60	14.4	33.3
60	48	5.76	11.5	17.3	39.9
70	56	6.72	13.4	20.2	46.6
80	64	7.68	15.4	23.1	53.2
90	72	8.64	17.3	25.9	59.9
100	80	9.60	19.2	28.8	66.5
110	88	10.6	21.1	31.7	73.2
125	100	12.0	24.0	36.0	83.1
150	120	14.4	28.8	43.2	99.8
175	140	16.8	33.6	50.4	116
200	160	19.2	38.4	57.6	133
225	180	21.6	43.2	64.8	150
250	200	24.0	48.0	72.1	166
300	240	28.8	57.6	86.5	200
350	280	33.6	67.2	101	233
400	320	38.4	76.8	115	266
450	360	43.2	86.4	130	299
500	400	48.0	96.0	144	333
600	480	57.6	115	173	399
700	560	67.2	134	202	466
800	640	76.8	154	231	532
1,000	800	96.0	192	288	665
1,200	960	115	230	346	798

kW figures are the continuous load at unity power factor. For a non-continuous load the full rating applies.

How the values are calculated

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Continuous load = breaker rating × 0.80
Minimum breaker = (continuous load × 1.25) + non-continuous load
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Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

The chart sizes a breaker from a load. It does not size the conductor, and the breaker must also protect the conductor at its ampacity. It does not apply to motor circuits, where NEC Article 430 sets protection by its own percentages, or to breakers in assemblies listed for operation at 100 percent of rating. It says nothing about interrupting rating, which is set by the available fault current.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/breaker-size-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.